

PRACTICAL-04

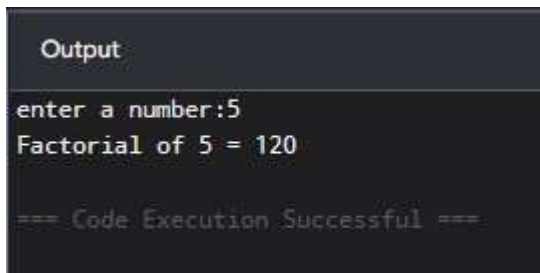
AIM: Implementation and Time analysis of factorial program using iterative and recursive method

1.) ITERATIVE METHOD

PROGRAM:

```
#include <iostream> using
namespace std; int main(){
int n;   int fact = 1;   cout
<< "enter a number:";   cin
>> n;   for(int i = 1; i <= n;
i++){       fact = fact * i;
    }
cout << "Factorial of " << n << " = " << fact;
return 0;
}
```

OUTPUT:



```
Output
enter a number:5
Factorial of 5 = 120

=== Code Execution Successful ===
```

2.) RECURSIVE METHOD:

PROGRAM:

```
#include <iostream>
```

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```
class factorial{
public:
```

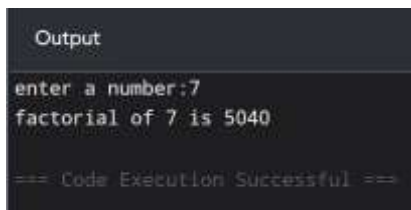
```
using namespace std;
```

```
int factr(int n){  
if(n == 0 || n == 1){  
return 1;  
} else{ int result =  
n*factr(n-1); return  
result;  
}  
}  
};
```

```
int main(){  
int n,result;  
factorial obj;
```

```
cout << "enter a number:"; cin>>n; result =  
obj.factr(n); cout << "factorial of " << n << "  
is " << result; return 0;  
}
```

OUTPUT:



Output

```
enter a number:7  
factorial of 7 is 5040  
=== Code Execution Successful ===
```

CONCLUSION:

Both iterative and recursive methods can be used to calculate the factorial of a number. The iterative method uses a loop and requires less memory, while the recursive method uses repeated function calls. Both have $O(n)$ time complexity, but the iterative method is generally more memory-efficient.

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